

Property tailorability for advanced CVI silicon carbide composites for fusion

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Received 8 April 2005; received in revised form 1 August 2005; accepted 1 August 2005

Available online 27 December 2005

Abstract

Chemically vapor infiltrated (CVI) silicon carbide (SiC) matrix composites with uni-directional and various two- and three-dimensional reinforcements by the near-stoichiometric SiC fibers or SiC/graphite hybrid fabrics were produced and evaluated for tensile, thermal and electrical properties. The parallel-serial approach models of these properties reasonably explained the experimental results. The experimental data and the model-based analysis suggested that, for the composite systems studied: (1) the strength properties are determined primarily by the volume fraction of longitudinal fibers, approximately following the theory that assumes the global load sharing; (2) presence of the axial fiber tows is the key factor in providing high thermal conductivity; (3) the maximum and minimum post-irradiation through-thickness conductivity of 10–15 W/m K at 800–1000 °C for 3D architecture and <5 W/m K at <800 °C for 2D architecture, respectively, and (4) the orthogonal 3D configurations of $x:y:z = 1:1:0.2$ to $1:1:0.5$ will provide the highest post-irradiation resistance to thermal stress due to a through-thickness heat flow.

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Keywords: Blanket material; SiC/SiC composite; Thermal conductivity; Electrical conductivity; Mechanical property; Thermal stress

1. Introduction

Silicon carbide (SiC) is an unique material that retains its strength and chemical stability at elevated temperatures and under intense neutron irradiation [1]. Additionally, SiC provides safety features when applied to fusion nuclear systems due to its low acti-

vation/low decay heat characteristics [2]. SiC fiber-reinforced SiC-matrix composites (SiC/SiC composites), which adds mechanical damage tolerance to inherently brittle monolithic SiC, are thus promising candidates for fusion blanket/first wall structures [3], and flow channel inserts [4]. Due to the complex role and design diversity of the breeding blankets of fusion reactors, the proposed application of SiC/SiC composites to various blanket designs demands a wide range of material properties [5]. Specifically, mechanical strength and thermal and electrical conductivities are

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the properties, which not only are of the primary concerns for fusion blanket applications but also are able to be anisotropically tailored in fibrous composites.

The objectives of this work are to determine the tailorability of anisotropic thermomechanical properties for chemically vapor-infiltrated (CVI) SiC/SiC composites and the trade-off between the mechanical and thermal properties. For these purposes, CVI SiC-matrix composites were fabricated employing the uni-directional (UD), common two-dimensional (2D), and three-dimensional (3D) reinforcements by near-stoichiometric SiC fibers and their mechanical, electrical- and thermal-transport properties were evaluated. The measured properties were analyzed using theoretical models and the prediction of the influences of neutron irradiation on temperature dependent composite properties were provided.

2. Experimental procedure

2.1. Materials

Various composites were produced through an isothermal/isobaric CVI process [6] as listed in Table 1. The composites were reinforced primarily by near-

stoichiometric SiC fibers, namely TyrannoTM-SA3 (Ube Industries, Ltd., Ube, Japan) or Hi-NicalonTM Type-S (Nippon Carbon Co., Tokyo, Japan). The Hybrid 3D composite incorporated ThornelTM P-120S pitch-based graphite fibers (Cytac Engineered Materials, Inc., Anaheim, CA) in the through-thickness or 'z' direction, whereas it used TyrannoTM-SA3 for the in-plane ('x' and 'y' directions) reinforcement. A single-layered pyrolytic carbon (PyC) or a multi-layered (ML) interphase was deposited [7].

2.2. Property evaluation

Mechanical properties were evaluated by tensile tests using rectangular bar specimens, in which the longitudinal direction was parallel to the x direction. The specimen dimensions were 40 mm (length) × 4 mm (width) × 2.3 mm (thickness) for 2D and 50 mm × 6 mm × 2.5 mm for 3D composites. The tensile test was conducted at room temperature following the general guidelines of ASTM C1275. Thermal diffusivity measurement was performed using a xenon lamp flash technique. Thermal conductivity was then calculated using the measured composite mass density and the composite specific heat estimated by the rule of mixture using the literature values for the constituents'

Table 1
List of materials studied

Material ID	Fiber	Reinforcement architecture	Fiber volume fractions (%)		Fiber/matrix interphase	Mass density (g/cm ³)	Porosity (%)
			x or y	z			
UD ML	HNLS ^a	UD ^d	38	0	ML ^{g,h}	2.3	26
UD PyC1	HNLS	UD	29	0	PyC ⁱ (500 nm)	2.6	19
UD PyC2	HNLS	UD	29	0	PyC (800 nm)	2.5	19
PW TySA	TySA ^b	2D PW ^e (0°/90°)	20	0	PyC (150 nm)	2.5	20
PW HNLS	HNLS	2D PW (0°/90°)	20	0	ML	2.6	18
SW HNLS	HNLS	5HSW ^f (0°/90°)	22	0	PyC (150 nm)	2.5	19
3D 1:1:1	TySA	3D Orthogonal, x:y:z = 1:1:1	15	15	PyC (150 nm)	2.8	11
3D 1:1:4	TySA	3D Orthogonal, x:y:z = 1:1:4	7.5	30	PyC (150 nm)	2.8	10
Hybrid 3D	TySA/P120S ^c	3D Hybrid ^g , x:y:z = 1:1:1.8	12	21	PyC (150 nm)	2.2	25

^a Hi-NicalonTM Type S.

^b TyrannoTM-SA3.

^c Thornel P-120S.

^d Uni-directional.

^e Plain-weave.

^f 5-Harness satin-weave.

^g See text.

^h (PyC/SiC) multi-layered interphase.

ⁱ Pyrolytic carbon.

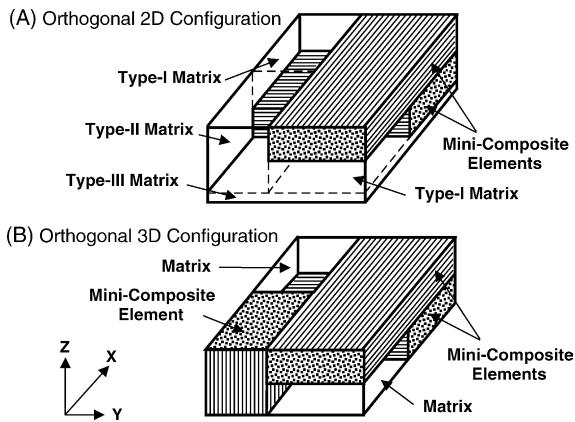


Fig. 1. Configurations of unit cells for orthogonal 2D (A) and 3D (B) architectures for the modeling of anisotropic modulus and transport properties of woven fabric CVI composites by parallel-serial approach.

specific heats. Electrical conductivity was measured using a four-point probe technique.

2.3. Model analysis

The composite elastic moduli and thermal conductivities were analyzed based on the parallel-serial approach (PSA) model proposed by Ishikawa et al. [8]. In this model, modulus and conductivity of the unit cell (shown in Fig. 1) in x , y , or z direction are calculated using a rule of mixture for multiple parallel and serial components, based on anisotropic moduli and conductivities of the ‘mini-composites’ (single tow + interphase + surrounding matrix) and the inter-tow matrix incorporating large pores. As for the 2D composites, the unit structure was approximated by the orthogonal 2D architecture as shown in Fig. 1A, incorporating three types of matrix elements. The Type-I and Type-II matrices are the inter-tow matrices, which include pores arising from the cross-over of tows and the fabric mesh, respectively, and a Type-III inter-laminar matrix, which includes large planar

inter-laminar pores. Each matrix element was given load/heat/electricity transfer efficiency factor, ε_w in direction w , which is determined by porosity (p), pore shape factor (F), and the angle between w and axis of symmetry for a planar pore (α) [9].

$$\varepsilon_w = (1 - p)^\beta; \quad \beta = \frac{1 - \cos^2 \alpha}{1 - F} + \frac{\cos^2 \alpha}{2F} \quad (1)$$

For matrix elements in the 3D composites, $p = 0.5$ and $F = 0.3$ were taken for most cases. For Type-I, II, III matrices in the 2D composites, $p = 0.4, 0.6, 0.5$ and $F = 0.15, 0.2, 0.1$ were used, respectively.

The constituent properties used in the analysis are summarized in Table 2. The geometric parameters were given based on the ceramographic examination. The effects of elevated temperatures and neutron irradiation on thermal conductivity were treated by modifying the thermal conductivity of each constituent in accordance with the linear thermal resistance model proposed by Snead et al. [10].

3. Results and discussion

3.1. Mechanical properties

The result of tensile test is summarized in Table 3. The 2D SiC/SiC composites exhibited ultimate tensile strength (UTS) and proportional limit stress (PLS) values comparable to literature results for similar materials [11,12]. The 3D composites exhibited significantly lower strength than the 2D SiC fiber composites. Under the assumed condition of global load sharing, the UTS is proportional to the product of longitudinal fiber volume fraction (V_f) and characteristic fiber strength [13]. On the other hand, the matrix cracking stress (σ_{mc}), which approximately corresponds to the PLS, should be roughly linearly related with V_f . Such relationships are seen in Fig. 2, where the measured UTS and PLS are plotted against

Table 2
Constituent properties used in calculation.

	Tyranno TM -SA3	Hi-Nicalon TM Type S	P-120S	Matrix SiC	PyC interphase
Elastic modulus (GPa)	420	400	840	450	12
Thermal cond. (20 °C) (W/m K)	64	18	640	70	5
Electrical cond. (20 °C) (S/m)	5	2	—	10	3×10^4

Table 3

Summary of experimentally measured (“Meas.”) and model-calculated (“Model”) properties

Material ID	UTS (MPa)	PLS (MPa)	Modulus (GPa)		Thermal conductivity (W/m K)				Electrical conductivity (S/m)	
	Meas.	Meas.	Meas.	Model	Meas. ^a	Model ^a	Meas. ^b	Model ^b	Meas. ^b	Model ^b
UD ML	249 (18)	249 (18)	306 (62)	321	n/m	–	n/m	–	200	208
UD PyC1	442 (120)	339 (12)	307 (8)	338	n/m	–	n/m	–	900	1160
UD PyC2	319 (56)	268 (37)	306 (51)	330	n/m	–	n/m	–	1700	1690
PW TySA	199 (2)	103 (21)	273 (23)	272	18.9 (0.3)	20.9	47.8 (1.3)	48.5	n/m	–
PW HNLS	n/m	n/m	n/m	286	18.6	16.1	n/m	33.9	n/m	–
SW HNLS	226 (50)	101 (9)	244 (6)	303	18.1 (0.3)	15.3	34.7 (1.3)	32.5	n/m	–
3D 1:1:1	102	53	227	225	46.1 (0.9)	46.3	47.4 (1.0)	44.1	n/m	–
3D 1:1:4	59 (13)	35 (2)	224 (5)	248	58.2 (1.8)	55.1	46.2 (1.4)	46.1	n/m	–
Hybrid 3D	98 (3)	47 (1)	95 (10) ^c	248	53.0 (13.5)	116.3	32.6 (0.5)	31.2	n/m	–

n/m: not measured. Numbers in parentheses are standard deviations.

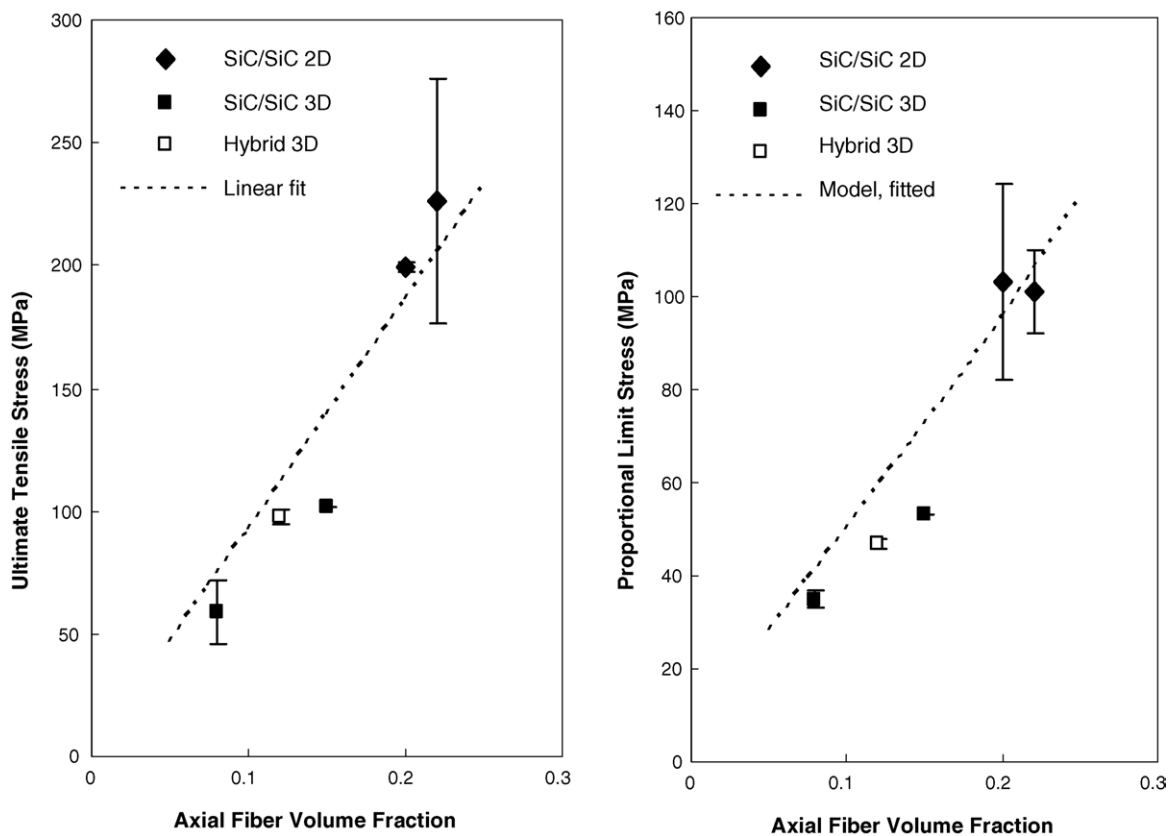
^a Through-thickness.^b In-plane.^c Matrix cracking occurred during processing.

Fig. 2. The influences of fiber volume fraction along the tensile axis on the ultimate tensile stress and proportional limit stress. Error bars indicate the standard deviation.

the longitudinal fiber volume fraction. The lower UTS for the 3D SiC/SiC composites are attributed to the reduced fiber strength due to the additional heat treatment to the 3D woven fabric. As shown in Table 3, the tensile modulus calculated using the model yielded a good agreement with the experimental result.

3.2. Thermal conductivity

Thermal conductivity measured along the in-plane and through-thickness directions at room temperature is summarized in Table 3. The data are also plotted in Fig. 3 as a function of the estimated contribution from the fibers along the heat transport axis to the composite thermal conductivity. The in-plane fiber contribution to through-thickness thermal conductivity in the 2D composite systems is due to the through-thickness component of the wavy fiber tows in 2D woven fabrics. The linear fit to the data points for 3D composites and in-plane thermal conductivity of 2D composites indicates the inclination of approximately unity and intersects the vertical axis at ~ 35 W/m K, in contrast to the 15–20 W/m K of through-thickness conductivity of the 2D composites. This implies a significant contribution of the matrix within and surrounding the axial fiber tows, in addition to the contribution from the axial fibers themselves, to the composite thermal conductivity. Therefore, the presence of the axial fiber tows is the key factor that enables a large heat conductance.

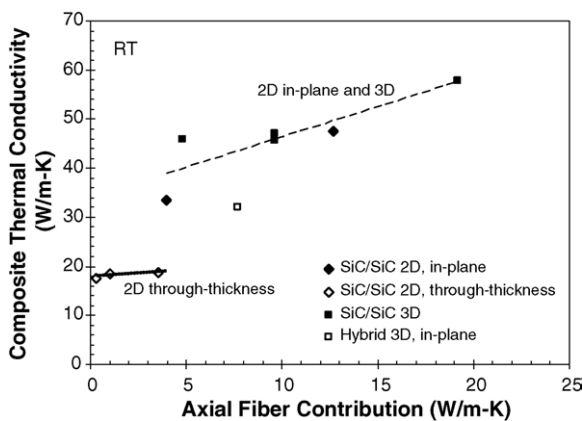


Fig. 3. Effect of heat conduction through axial fibers on composite thermal conductivity at room temperature. The dotted line represents a linear fit to data points for SiC/SiC composites in one of fiber directions.

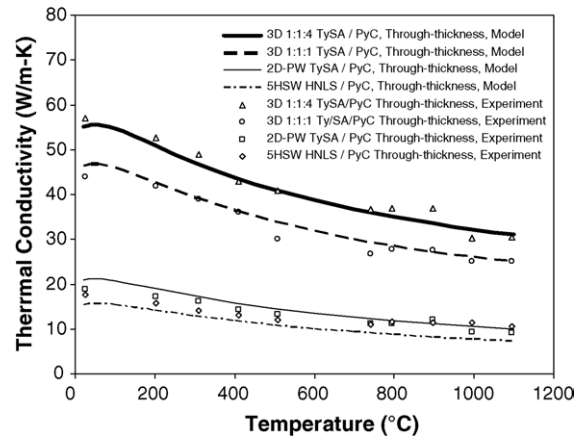


Fig. 4. Temperature-dependent thermal conductivity of various CVI SiC/SiC composites: comparison of model-prediction and experimental data.

The model-prediction showed a good agreement between the calculated and experimental thermal conductivity in both the in-plane and through-thickness directions, as shown in Table 3. The geometric parameters used for thermal conductivity are identical with those used for the elastic modulus calculation. It was assumed that these results validate the applicability of this approach to the approximate estimation of various transport properties in the present composite systems.

In Fig. 4, the calculated temperature dependence of thermal conductivity for selected composites is presented along with experimental data [14]. For this calculation, linearly temperature dependent reciprocal thermal diffusivity was assumed for both matrix and fibers, based on the thermal conductivity data for chemically vapor-deposited SiC [14], and the coefficient of thermal resistance was taken to be $0.01 \text{ s/cm}^2 \text{ K}$ to fit the experimental data in Fig. 4.

Neutron irradiation is known to further increase thermal resistance for SiC, which initially increase with fluence and tends to saturate by a few dpa of neutron fluence. The saturated defect thermal resistance is linearly dependent on the irradiation temperature at 300–800 °C [15] and the data could be fitted to the equation below:

$$\frac{1}{K_{\text{rd}}} = 0.119 - 1.13 \times 10^{-4} T(^{\circ}\text{C}) \quad (2)$$

Fig. 5 presents the post-irradiation thermal conductivity at the irradiation temperature calculated using Eq. (2). The positive temperature dependence of the post-

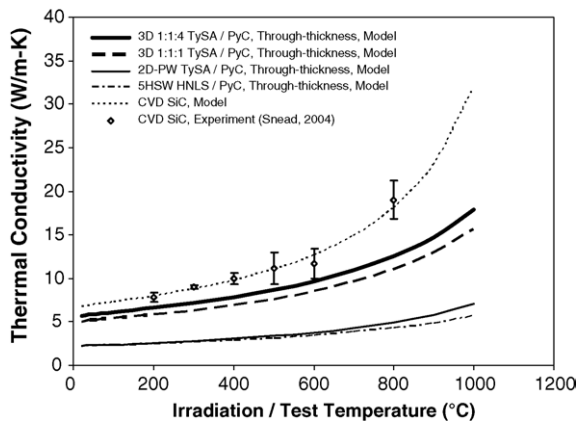


Fig. 5. Post-irradiation thermal conductivity of CVI SiC/SiC composites as a function of temperature predicted by the model. The temperature at which thermal conductivity was calculated is equal to the irradiation temperature.

irradiation thermal conductivity is due to the strong contribution from the defect thermal resistance, in contrast to the negative temperature dependence for the non-irradiated materials. The model prediction suggests that the post-irradiation through-thickness thermal conductivity of 10–15 W/m K is feasible for high-heat flux applications at 800–1000 °C, which is the temperature range of primary interest for power reactor design studies [5], when through-thickness fibers are incorporated in CVI SiC-matrix composites. Meanwhile, 2D laminated composites are more appropriate when the through-thickness thermal conductivity <5 W/m K is desired under irradiation at <800 °C, which is of interest for the flow channel insert application [4]. Substantial increase in through-thickness thermal conductivity from this upper limit will require the incorporation of heat conductive media within the composites [16]. Substantial decrease may also be achieved

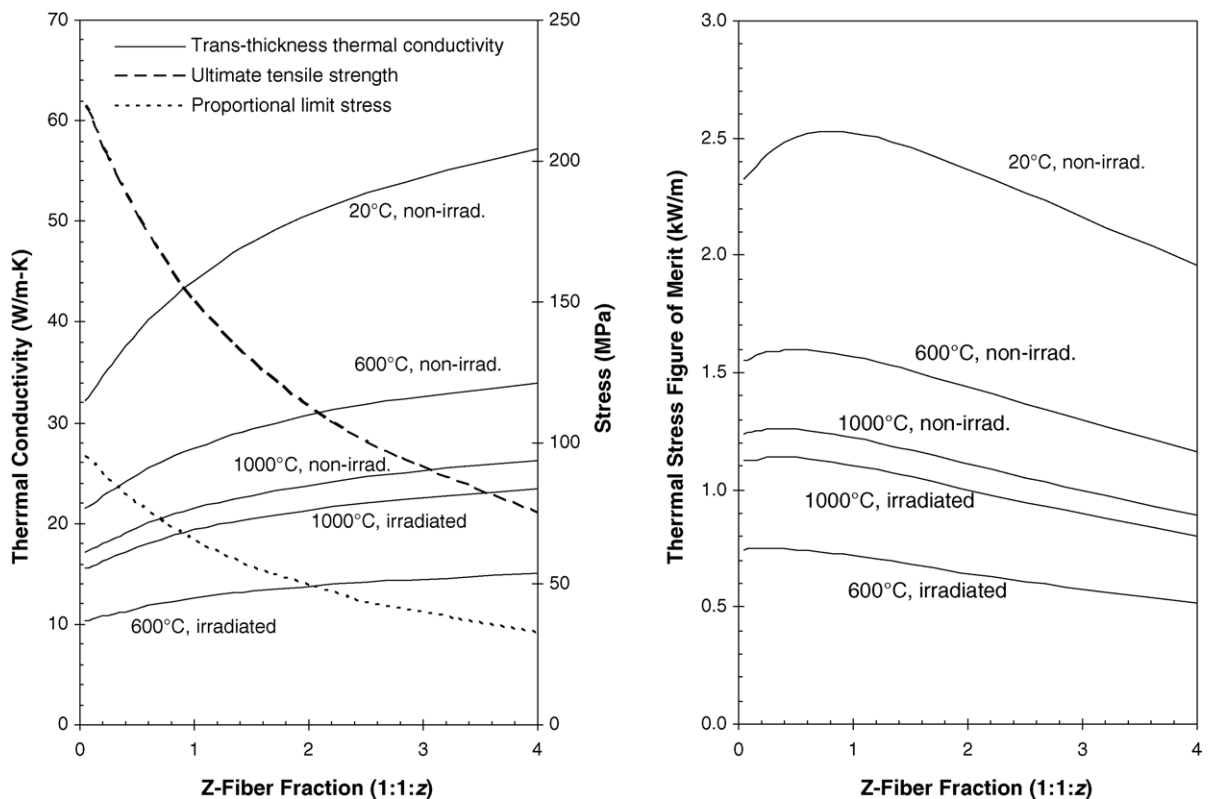


Fig. 6. The predicted influences of z -fiber volume fraction in 1:1:z orthogonal 3D configuration on non-irradiated and irradiated through-thickness thermal conductivity, in-plane tensile properties, and thermal stress figure of merit for TyrannoTM-SA CVI SiC-matrix composites.

by incorporating thermally insulating media, including pores, or by impurity doping to the SiC-matrix.

3.3. Optimizing the 3D configuration

For high through-thickness heat flux applications such as blanket first walls, thermal stress arising from the temperature gradient across the wall thickness will be a major design issue that limits the operating condition of specific systems [5]. The general resistance of materials to thermal stress is conveniently indexed by a thermal stress figure of merit (M) [17]:

$$M = \frac{\sigma_a K_{th}(1 - \nu)}{\alpha E} \quad (3)$$

using the allowable design stress (σ_a), through-thickness thermal conductivity (K_{th}), Poisson's ratio (ν), coefficient of in-plane thermal expansion (α), and in-plane elastic modulus (E). In Fig. 6, the predicted influences of z -fiber fraction in 1:1:1 configuration on the through-thickness thermal conductivity, tensile strength properties, and the thermal stress figure of merit for 3D SiC/SiC composites are presented. The PLS was used as the allowable stress. The influences of elevated temperature and neutron irradiation on the mechanical properties were neglected since they should not greatly alter the z -fiber fraction dependence and the magnitude of the influences was reported to be minor [3,18]. The result indicates that the highest resistance to thermal stress occurs at 1:1:0.5–1.0 at room temperature before irradiation, but the z -fiber fraction that gives the peak M values shifts to 1:1:0.2–0.5 at elevated temperatures and/or after irradiation. Also, the M value appeared insensitive to the z -fiber fraction in this range.

4. Conclusions

CVI SiC-matrix composites with various UD, 2D, and 3D reinforcements by near-stoichiometric SiC fibers or SiC/graphite hybrid fabrics were produced and evaluated for tensile properties and thermal conductivity. The models of tensile properties, elastic modulus, and thermal conductivity reasonably explained the experimental results obtained. The following implications were obtained based on the experimental data and the model-based analysis:

- (1) The UTS and PLS of the composites are determined primarily by the volume fraction of longitudinal fibers, provided that the other constitutive properties are similar, approximately following the theory of global load sharing.
- (2) Thermal conductivity of the composite systems studied is determined by the significant contributions from the axial fibers (if present), matrix within and surrounding the axial fiber tows (if present), and the remaining constituents of the composite. Presence of the axial fiber tows is the key factor in determining the thermal conductivity.
- (3) Effect of neutron irradiation on temperature dependent thermal conductivity of the composites was predicted using the linear thermal resistance model and the temperature dependent defect thermal resistance data for the composites' constituents. The prediction suggests the maximum and minimum post-irradiation through-thickness conductivity of 10–15 W/mK at 800–1000 °C for 3D architecture and <5 W/mK at <800 °C for 2D architecture, respectively, for CVI SiC-matrix composites in the porosity range studied.
- (4) The highest resistance to thermal stress at elevated temperatures and/or after neutron irradiation is likely to occur at $x:y:z = 1:1:0.2$ – $1:1:0.5$ configurations for the orthogonal 3D composites.

Acknowledgements

Thanks are extended to Drs. H.-T. Lin and S.J. Zinkle for reviewing the manuscript. This work was sponsored by the Office of Fusion Energy Sciences, US Department of Energy under contract DE-AC05-00OR22725 with UT-Battelle, LLC, and the 'JUPITER-II' US-Department of Energy/Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT) collaboration for fusion blanket and material systems research.

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